(KTXFI1EBNF) Physics I. Practice

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April 13, 2026

Exercises I

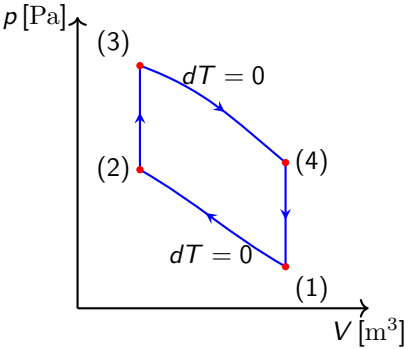
- **Thermodynamics I:** Temperature scales, thermal expansion, thermometers. State variables, ideal gas, state changes, ideal gas equation of state, Boyle's law, Gay-Lussac's laws, universal (combined) gas law, heat, specific heat, molar heat, heat capacity.
- **Thermodynamics II:** Expansion work, internal energy, the first law of thermodynamics, forms of the first law in different state changes.
- **Thermodynamics III:** Cyclic processes, Carnot cycle, Carnot's theorem, reduced heat, entropy, the second law of thermodynamics.
- **Thermodynamics IV:** Kinetic theory of gases, molecular thermodynamics. Distributions, distribution functions. Elementary statistics.
- **Electrostatics I.:** Electric charge: Fundamental electrical phenomena, electric charge and matter, Coulomb's law (force between two point charges, unit of electric charge, permittivity of vacuum).

Exercises II

- **Electrostatics II.:** The electric field: The electrostatic field (electric field strength, electric field lines). Determination of electric field strength. Gauss's law (flux of the electric field). Electric field of a point charge.

Exercises III

- Problems:
  11. The ideal cycle of an externally heated heat engine, shown in the diagram, differs from the Otto engine cycle in that it contains isothermal processes instead of adiabatic ones.



What is the efficiency of the cycle if $p_3 = 4p_2$, the compression ratio is 6, and the cycle is performed with a two-atomic (diatomic) gas? What is the efficiency of the Carnot cycle taking place between the same temperature limits?

Exercises IV

The ideal cycle of an externally heated heat engine differs from the Otto engine cycle in that it contains isothermal processes instead of adiabatic ones. Calculate:

a) The efficiency (η) of the cycle if $p_3 = 4p_2$, the compression ratio is $r = 6$, and the gas is diatomic.

b) The efficiency of a Carnot cycle (η_{Carnot}) operating between the same temperature limits.

- Pressure ratio: $p_3 = 4p_2$
- Compression ratio: $r = \frac{V_1}{V_2} = 6 \implies V_1 = 6V_2$
- Gas type: Diatomic (Two-atomic) $\implies C_V = \frac{5}{2}R$

Exercises V

Efficiency of the Cycle

State (1) to State (2): Isothermal Compression ($T_1 = T_2$) According to Boyle-Mariotte's law:

$$p_1 V_1 = p_2 V_2 \implies p_2 = p_1 \frac{V_1}{V_2} = p_1 \frac{6V_2}{V_2} = 6p_1$$

The heat released during this process:

$$Q_{released,12} = \frac{m}{M} R T_1 \ln\left(\frac{V_2}{V_1}\right) = \frac{m}{M} R T_1 \ln\left(\frac{1}{6}\right)$$

State (2) to State (3): Isochoric Heating ($V_2 = V_3$) According to Gay-Lussac's II law:

$$\frac{p_2}{T_2} = \frac{p_3}{T_3} \implies \frac{T_3}{T_2} = \frac{p_3}{p_2} = 4 \implies T_3 = 4T_2 = 4T_1$$

Exercises VI

The heat absorbed during this process:

$$Q_{absorbed,23} = C_V \frac{m}{M}(T_3 - T_2) = \frac{5}{2}R\frac{m}{M}(4T_1 - T_1) = \frac{15}{2}R\frac{m}{M}T_1$$

State (3) to State (4): Isothermal Expansion ($T_3 = T_4$) Since $T_3 = T_4 = 4T_1$ and $V_4 = V_1, V_3 = V_2$:

$$Q_{absorbed,34} = \frac{m}{M}RT_3 \ln\left(\frac{V_4}{V_3}\right) = \frac{m}{M}R(4T_1)\ln\left(\frac{6V_2}{V_2}\right) = \frac{m}{M}RT_1 \cdot 4\ln 6$$

State (4) to State (1): Isochoric Cooling ($V_4 = V_1$) The heat released during this process:

$$Q_{released,41} = C_V \frac{m}{M}(T_1 - T_4) = \frac{5}{2}R\frac{m}{M}(T_1 - 4T_1) = -\frac{15}{2}R\frac{m}{M}T_1$$

Exercises VII

Efficiency Calculation: The efficiency is defined as $\eta = \frac{\sum Q}{Q_{absorbed,total}}$. The total absorbed heat is $Q_{absorbed} = Q_{23} + Q_{34}$. The net heat is $\sum Q = Q_{12} + Q_{23} + Q_{34} + Q_{41}$.

$$\eta = \frac{\frac{m}{M}RT_1 \ln(\frac{1}{6}) + \frac{15}{2}R\frac{m}{M}T_1 + \frac{m}{M}RT_1 \cdot 4\ln 6 - \frac{15}{2}R\frac{m}{M}T_1}{\frac{15}{2}R\frac{m}{M}T_1 + \frac{m}{M}RT_1 \cdot 4\ln 6}$$

Simplifying by $\frac{m}{M}RT_1$:

$$\eta = \frac{-\ln 6 + 4\ln 6}{7.5 + 4\ln 6} = \frac{3\ln 6}{7.5 + 4\ln 6} \approx \frac{5.375}{14.667} \approx 0.3665$$

$$\eta = \mathbf{36.65}\%$$

Exercises VIII

Carnot Efficiency The Carnot efficiency between $T_H = 4T_1$ and $T_L = T_1$:

$$\eta_{Carnot} = \frac{T_H - T_L}{T_H} = \frac{4T_1 - T_1}{4T_1} = \frac{3}{4} = 0.75$$

$$\eta_{\textbf{Carnot}} = \mathbf{75}\%$$

The End

Thank you for your attention!